LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Tyres and tubes tyres variation of tyre pressure with all up weight, creep and creep marking on wheels Tyre and brake care
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Tyres and tubes tyres variation of tyre pressure with all up weight, creep and creep marking on wheels Tyre and brake care
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Inspection of tyre pressure gauges and identification of creep marks on wheel assemblies; Safety: Ensure aircraft is chocked and brakes set; Supervision: Instructor monitors gauge reading accuracy and safety protocols.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft landing gear tyres?
(b) How does aircraft all-up weight influence tyre pressure requirements?
(c) What visual indicators are used to detect wheel creep?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Typical alighting gear construction • Tyre pressure variation with all up weight • Creep marking identification on wheels • Tyre and brake care procedures	L & Ds	PPT, IB	15 min
(b)	• Impact of weight distribution on pressure • Consequences of incorrect tyre pressure • Maintenance schedules for tyres and brakes • Inspection techniques for creep marks	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of pressure checking • Identification of creep marks on training aid • Review of tyre care best practices • Q&A session on maintenance protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how increasing all-up weight affects the required tyre pressure.
(b) Describe the significance of creep marks on aircraft wheels.
(c) List two key maintenance steps for ensuring tyre and brake longevity.

12. Home Assignment and Follow Up	Prepare a brief note on the basic design requirements of a pneumatic system to be discussed in the next lesson (LP 18).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Pneumatic system basic design requirements, brief description, advantages and disadvantages of a typical pneumatic system.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min